

DANESH AURIGA

RoboCup Junior Rescue

Simulation 2026

Country: Iran

“Minimizing the Golden Time
in Rescue Operations through
Autonomous Intelligence.”

TEAM IDENTITY & INSPIRATION

Every great rescue mission starts with a spark of inspiration. Formed in September 2025, Team “Danesh Auriga” embodies the perfect fusion of intellect and motion. We aren't just writing Python scripts; we are engineering a highly autonomous digital lifeline, designed to accelerate victim identification in the most unforgiving, hazardous environments.



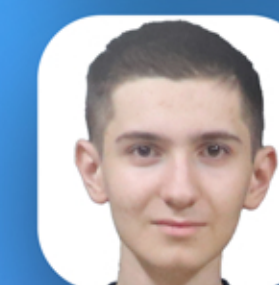
First Place at the 2026 RoboCup National Competitions (Iran Open)

MEMBERS

Artin Taleei
Image Processing & AI



Ryan Feizbakhsh
Mapping



Radin Mohammadian Asl
Navigation & Decision Making



Amirsam Eftekharinia
Image Processing & AI



HARDWARE ARCHITECTURE & SENSORS

Our hardware in Webots is optimized for off-grid navigation, robustness, and precise perception.

Vision System (3x Cameras) :

Top-down multi-camera setup removes close-range occlusion and expands coverage, enabling reliable victim detection (including fake/real via 3D features) and tile/color recognition—eliminating the need for color sensor and short range distance sensors.

LiDAR:

Provides high-precision structural data for reliable wall detection and occupancy grid mapping.

Central GPS + InertialUnit :

Co-located at the robot's center of mass for stable localization and accurate orientation. This reduces sensor fusion error and improves mapping and motion stability.

Encoders :

Used for accurate odometry and short-term motion estimation. The combination of encoders, GPS, and LiDAR makes an accelerometer unnecessary.

Distance Sensor (Auxiliary) :

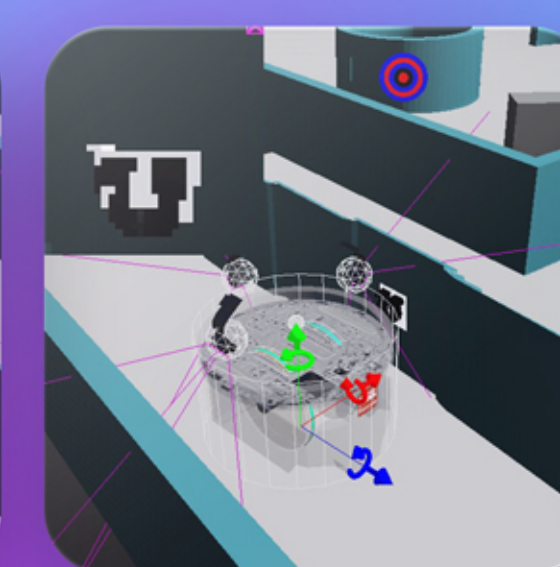
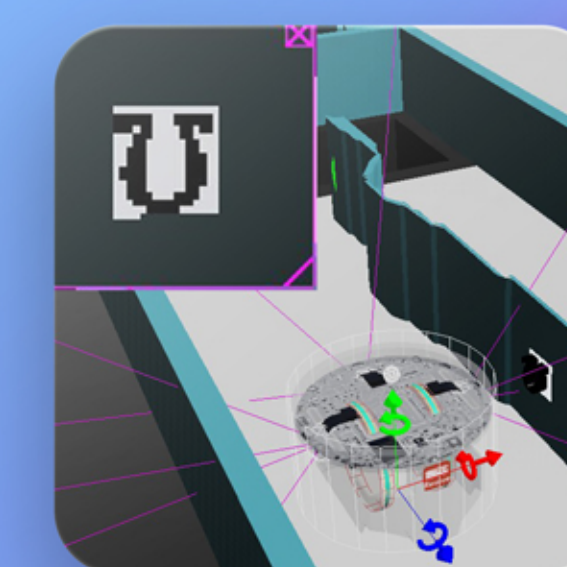
We integrated this sensor using remaining resource capacity.

Sensor Choice :

InertialUnit selected over Gyro based on testing.

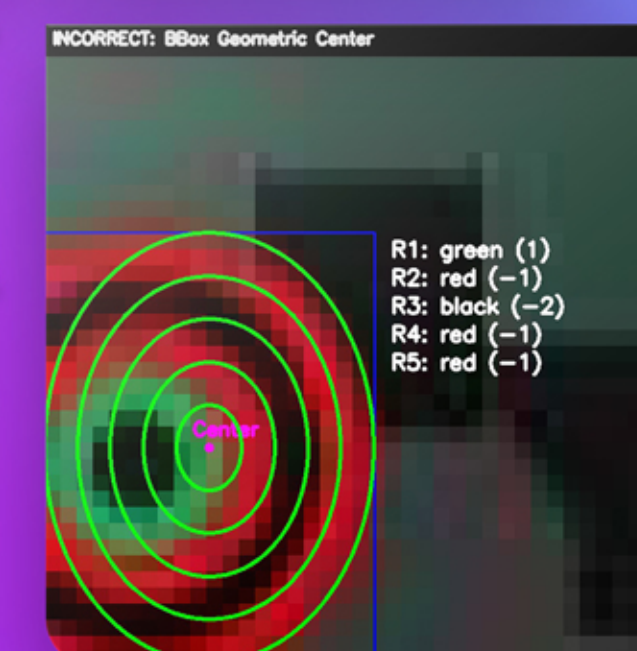


INOVATIONS



By increasing the camera height and using a top-down viewing angle, as you can see, the difference between fake victims and real ones is obvious due to the depth of fake victims

Using keypoints along-side object detection helps us accurately find the center of the circle, and subsequently the rings, even in scenarios where the cognitive target is not fully visible in the image or suffers from perspective distortion.



SOFTWARE ARCHITECTURE

At the core lies a Python Finite State Machine (FSM) orchestrating exploration, victim identification, and rescue – predictive and analytical, not merely reactive.

Occupancy Grid Mapping & Localization :

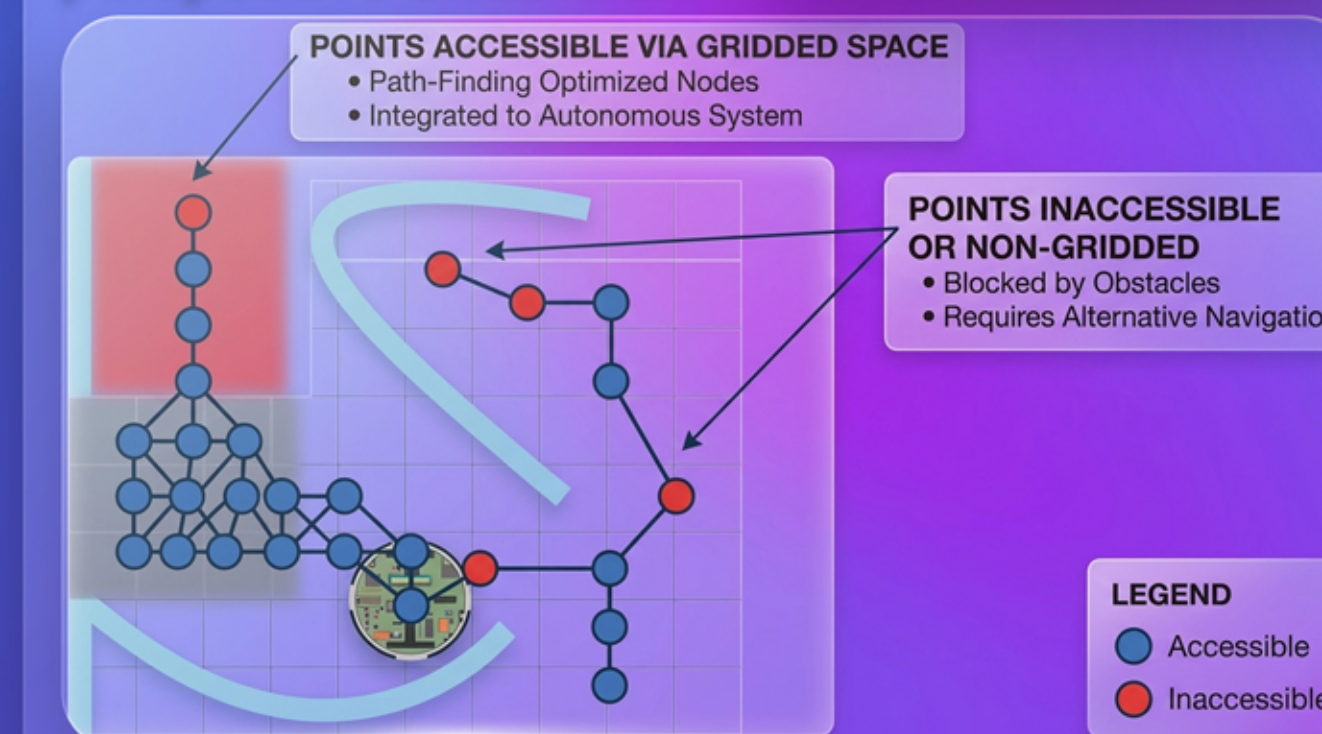
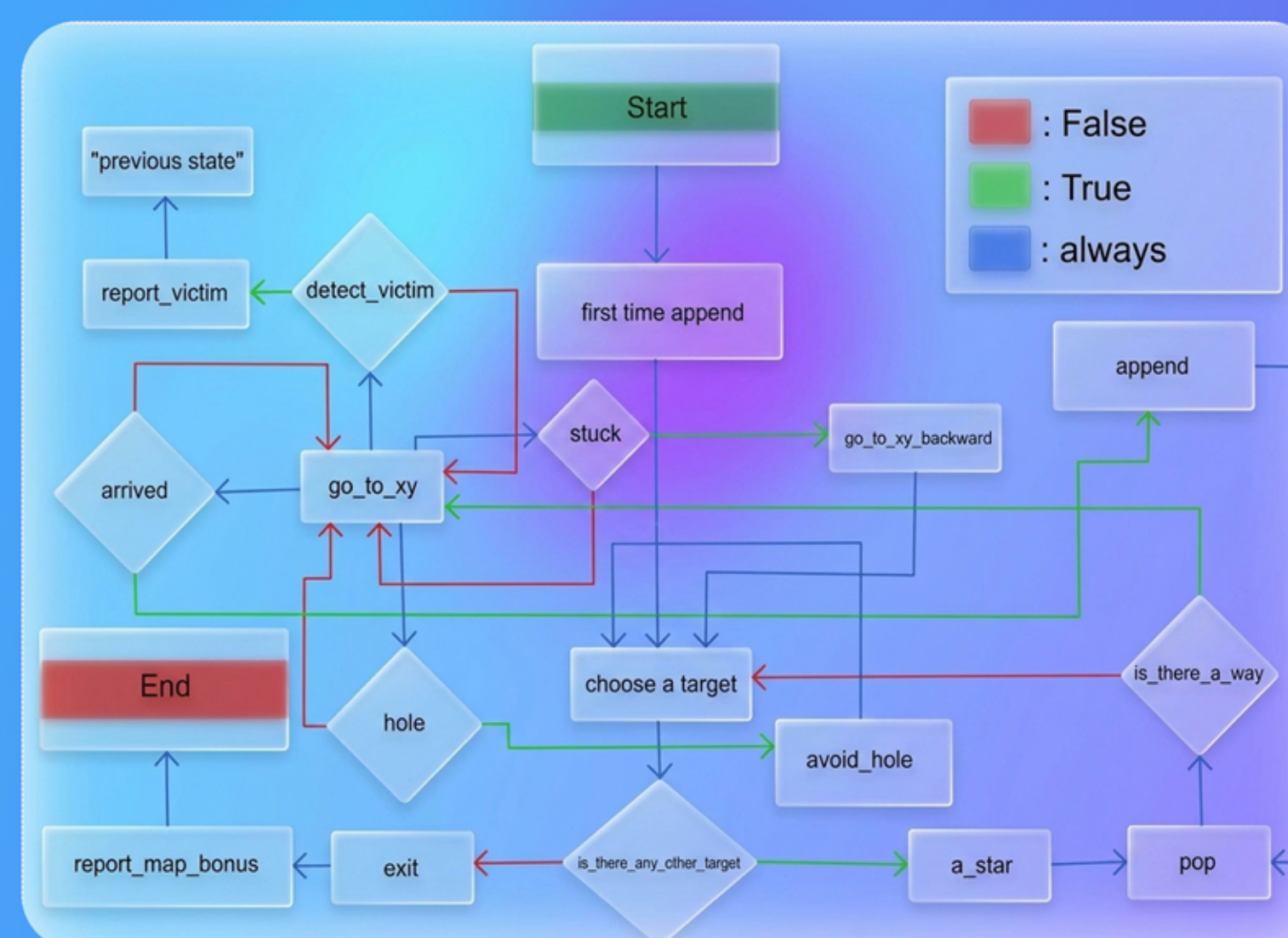
We abandoned grid-based navigation for graph based navigation (NetworkX). The robot logs tile coordinates, walls, hazardous zones (holes, swamps), and colored tiles in real time. Testing shows an average of 88% reconstruction accuracy under simulated GPS noise.

Precision Path & Motion Planning (A* Integration)*

Efficient navigation is critical. We implemented A* (Euclidean heuristic) with node traversability validation – automatically avoiding walls and hazards. A proportional controller ensures smooth, off-grid motion.

Multi-Modal Victim Detection (Vision & Depth) :

Our dual-layered pipeline fuses RGB and LiDAR: YOLOv8-pose detects structural keypoints ($\Phi/\Psi/\Omega$), while a custom circle detector analyses colour/edges for cognitive targets. A persistent victim-off list prevents duplicate reports.



Enhanced graph-based navigation to expand map coverage and enable exploration of previously unreachable areas.

